

Mainul Islam

Research Experience, Skills, and Interests

Department of EEE, Islamic University of Technology, Gazipur, Bangladesh | mainulislam@iut-dhaka.edu

This summary spans seven publications (five published, two under review) and hands-on projects across control, machine learning, computer vision, embedded systems, autonomous systems, and robotics.

Research Interests

Building on my experience across control, machine learning, and embedded robotics, I am interested in pursuing future research on:

- Learning-based and adaptive control (including reinforcement learning) for autonomous and multi-agent robotic systems.
- Robust perception, sensor fusion, and state estimation for reliable autonomy in unstructured, real-world environments.
- AI-enabled digital twins and control co-design that couple simulation, learning, and physical-system design.
- Efficient, interpretable, and trustworthy machine learning for edge and embedded deployment in safety-critical applications.
- Real-time, fault-tolerant embedded and cyber-physical systems for field robotics and IoT.

Control Systems, Autonomous Systems & Robotics

- **MonoSIM** (*published, IEEE ICARA*): Open-source Software-in-the-Loop framework for Ackermann-steered autonomous vehicles; implemented and benchmarked Model Predictive Control (MPC) and PID against a monocular-vision lane-detection pipeline — the simulate–design–validate loop central to control co-design and digital twins.
- **Speed Synchronization of a Dual-BLDC Tracked Vehicle** (*under review*): GA-tuned multi-rate cross-coupled controller for precise speed synchronization of a differential-drive tracked platform.
- **Autonomous Ground Vehicle — CYBERFLEET** (project): ROS2 autonomy stack for a 3-robot warehouse fleet — a from-scratch 5-state EKF fusing AprilTag vision with wheel odometry, space-time A* planning, trajectory tracking, and LiDAR obstacle avoidance.
- **Man-Transportable EOD ROV (JS2.0)** (project): System architecture for an 11-motor, dual-CAN robot — closed-form 7-DOF inverse kinematics (verified with MoveIt 2), redundant wired/wireless control, and layered fail-safe logic.
- **EOD ROVs — Jontro Soinik 1.0 & Smart Soinik 1.0** (delivered): Real-time embedded firmware and addressable-UART modular control for field-deployed robots operating under the Bangladesh Army and a UN Peacekeeping Mission.
- **Autonomous Target-Tracking Drone** (project): Onboard-vision detection and tracking (YOLO on Raspberry Pi 5; DeepSORT/KCF) integrated with the flight controller.
- **Surveillance Drone (SD15-N)** (project) & **Fixed-Wing VTOL UAV** (project): Long-endurance-UAV flight-time optimization (drag/thrust/battery), and a prototype study of aerodynamic stability, control allocation, and VTOL-to-fixed-wing transition.

Machine Learning, Computer Vision & Deep Learning

- **Bangla License Plate Recognition** (*published, IEEE ICAD*): Two-stage adaptive YOLOv8 detector paired with a ViT + BanglaBERT vision-language OCR pipeline, validated on real low-light CCTV data.
- **Solar Panel Fault Detection** (*published, Solar Energy, IF 6.6*): Hybrid RefineU-Net segmentation and Mo-

bileNetV3 classification for multi-class PV fault detection, with LIME-based explainable AI for interpretability.

- **Long-Term Energy Demand Forecasting** (*published, IEEE ICECIE*): Benchmarked Random Forest, KNN, XGBoost, and LightGBM on a million-sample utility dataset for long-horizon demand prediction.

Embedded Systems, Power Electronics & Hardware

- **Multi-node Real-Time Control** (*published, IEEE ICCIT*): Master-slave RS485/UART architecture across 11+ STM32 microcontrollers with DMA and CRC for reliable, low-latency distributed control.
- **Topological Survey of Zeta Power Converters** (*under review*): Comparative study of Zeta converter topologies, from classical designs to emerging architectures.
- **Toll-Booth Automation** (project): Industrial multi-lane STM32 controller — table-driven FSM firmware, DMA-UART and WIZnet W5500 TCP networking, and three custom PCBs with galvanic isolation.
- **Offline Hardware CryptoWallet** (project): Air-gapped embedded wallet — system state machine, touch-screen UI, BIP-39 key generation, and QR-based offline transaction signing.
- **Undergraduate & team hardware**: 8-bit SAP-1 computer; LiPo Battery Management System; pulse-width measurement/generation on the AT89S52; and IUT Mars Rover (Project Altair, Team Avijatrik) — PCB design, hardware integration, and debugging for international rover challenges.

Technical Skills

Python (PyTorch, scikit-learn), C/C++ (embedded, STM32), ROS2, real-time embedded control (UART, RS485, CAN bus, Modbus, DMA), classical estimation and control (EKF, PID, MPC), computer vision, PCB design, Git, and end-to-end deployment of vision and forecasting models from data collection through publication.